KinetiQ

Functional Requirements

Milestone 2 · Lifecycle Architecture · PEI 2026/2027, Universidade de Aveiro

KinetiQ functional requirements, aligned with the final MS2 architecture (revised on 31 Oct 2026), the domain model and the four system actors: Fleet Operator, Maintenance Technician, Fleet Analyst and Rider. The Fleet Operator also administers the system.

Telemetry Ingestion and Processing

ID	Requirement	Description
FR01	Asynchronous telemetry ingestion	The system must receive telemetry from the sources (ESP32 sensors and simulators) through the MQTT Broker. The MQTT Source Connector forwards it to Kafka (topic `telemetry.device.raw`), which feeds real-time processing.
FR02	Validation and normalisation	The Telemetry Processing Service must validate each message against the agreed schema and normalise it. Valid messages go to the `telemetry.device.processed` topic; malformed ones are sent to a Dead-Letter Queue (DLQ) for auditing.
FR03	Time-series persistence	Processed telemetry must be stored in InfluxDB, asynchronously from Kafka, through the InfluxDB Sink Connector.
FR04	Real-time broadcast	The system must push validated telemetry and events in real time to connected clients (web back-office and mobile app) through the WebSocket Service and the API Gateway (WSS).

Asset and Fleet Lifecycle Management

ID	Requirement	Description
FR05	Bike registration	The Fleet Operator must be able to register, update and retire bikes, linking physical identifiers (frame number, QR code) and logical ones (IoT device, model, acquisition date).
FR06	State machine	The system must manage each bike's lifecycle with a state machine (Available, In Use, In Maintenance, Retired), preventing invalid transitions. Every transition is published to the `fleet.events.status` topic.
FR07	History and traceability	The system must keep the full history of each bike, including accumulated mileage, routes and reported faults.
FR08	Maintenance work	The Maintenance Technician must be able to consult maintenance tickets, claim a ticket, record the work done and move the bike from In Maintenance to Available, closing the ticket. **To be confirmed:** whether the bike enters In Maintenance when the ticket is created or when the technician claims it.
FR09	Aggregated fleet metrics	The system must continuously compute fleet indicators: availability ratio, number of active bikes, average battery consumption and hours of use.
FR26	Geographic zone management	The Fleet Operator must be able to create, edit and remove geographic zones (operation area, restricted zone and speed-limit zone), stored in PostgreSQL/PostGIS. Changes are published to Kafka so that the Telemetry Processing Service takes them into account.

Monitoring, Alerts and Predictive Maintenance

ID	Requirement	Description
FR10	Threshold alerts	The system must raise alerts when configurable thresholds are reached: low battery (e.g. below 15%), prolonged loss of connection, leaving an authorised zone and speeding. **To be confirmed:** whether the speed limit is set per zone or as a global threshold (FR25); the domain model does not yet have a speeding alert type.
FR11	Alert routing	Critical alerts must reach the Maintenance Technician's view in real time through the WebSocket Service, and the Fleet Management Service must open a maintenance ticket for the bike.
FR12	ML anomaly detection	The ML module must analyse processed telemetry (topic `telemetry.device.processed`) and history to detect abnormal behaviour and wear, and raise alerts. **To be confirmed:** ML scope for the MVP: anomaly detection (FR12), demand forecasting (FR14) or both.

Analytics and Sustainability

ID	Requirement	Description
FR13	CO_2 savings	The system must compute CO_2 savings per trip and accumulated for the fleet and for each Rider, compared with average emission factors of equivalent polluting transport. **To be confirmed:** which component runs the calculation (the diagram only has the ML and BI modules).
FR14	Demand forecasting	The ML module must forecast high-demand areas to support fleet redistribution decisions. **To be confirmed:** whether it is part of the MVP (see FR12).
FR15	BI reports	The BI module must produce reports and aggregated views on fleet usage and status, available through the Core API.
FR16	Extensible API	The system must expose a documented REST API that serves the web and mobile clients and lets new value-added modules plug in through Kafka.

Web Back-Office

ID	Requirement	Description
FR17	Real-time map	The back-office must show an interactive map with the location and current status of the whole fleet. **To be confirmed:** the team has flagged this requirement as uncertain.
FR18	KPI and sustainability dashboard	The Fleet Operator must have a dashboard with the fleet KPIs (FR09) and total CO_2 savings (FR13).
FR19	History search and filters	The back-office must allow searching and filtering the trip, telemetry and status history of any bike by identifier and date range.

Rider Mobile App

ID	Requirement	Description
FR20	Bike association	The Rider must be able to associate with a bike in the mobile app by scanning its QR code; the bike moves to In Use.
FR21	Live trip	The mobile app must show the current trip's speed, battery level and estimated range in real time.
FR22	Personal history and CO_2	The Rider must be able to view their trip history, distance travelled and accumulated CO_2 savings.
FR23	Fault reporting	The Rider must be able to report a bike fault through the mobile app.

Administration and Access Control

ID	Requirement	Description
FR24	Authentication and roles (RBAC)	Authentication and permissions are managed centrally by Keycloak, with the roles Operator, Technician, Analyst and Rider. The API Gateway validates the access token on every request.
FR25	System configuration	The Fleet Operator must be able to manage users and roles, configure alert thresholds and manage integration keys.

Open points

Points not yet decided by the team that affect these requirements.

Requirement	Open point
FR08, FR11	When the bike enters In Maintenance: when the ticket is created or when the technician claims it.
FR10	Speed limit per zone or global; the domain model lacks a speeding alert type.
FR10, FR11	Alerts topic name: `fleet.alerts.system` or `maintenance-alerts` (used in the MS2 slides).
FR12, FR14	ML scope for the MVP: anomaly detection, demand forecasting or both.
FR13	Component responsible for the CO_2 calculation.
FR17	Requirement flagged as uncertain by the team.